A Latent Oscillator Measurement Model for Simulating Emotional-Expression Score Dynamics in Video

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Abstract

Automated vision models turn video into dense time series of model-derived emotional-expression scores. Before these scores are interpreted dynamically, researchers need to know whether analytic methods can separate latent expressive organization from the measurement process that produces the observed channels. We introduce the Latent Oscillator Measurement Model (LOMM), a controlled simulation framework that generates latent oscillator trajectories, gates their activity over time, and maps them through block loadings into noisy score channels. The generator varies latent dimensionality, feedback regime, activity pattern, loading strength, and measurement error. We calibrated LOMM against empirical video-score panels using targets for amplitude dispersion, local movement, temporal persistence, directional-change complexity, spectral complexity, and effective multivariate dimensionality. Empirical realism was evaluated with a softmax-linked bounded score panel, whereas the recovery simulations used pre-link linear block-measurement panels to isolate dimensional recovery under a known measurement structure. The calibrated generator reproduced several local temporal and dimensional features, while retaining an amplitude-dispersion mismatch. We then evaluated dimensional recovery in two designs: a broad DynEGA grid and a matched comparison of DynEGA, GraphicalVAR, and GIMME using common community-detection readouts. DynEGA showed the most consistent dimensional and block recovery in these LOMM simulation designs and completed the broad grid. GraphicalVAR often recovered the generating dimensionality but was slower and more sensitive to weak measurement conditions. GIMME served as an indirect feasibility-stress comparator with substantial runtime and completion limits in this design. LOMM provides a reusable framework for testing whether dynamic methods recover latent organization, tolerate measurement error, and scale to video-derived behavioral time series before substantive interpretation.

Introduction

Video is increasingly used as a source of intensive behavioral data because it preserves the temporal ordering of visible action and can be analyzed at a resolution that manual frame-by-frame coding cannot sustain. Deep-learning systems can now convert frames or short video segments into dense time series of model-derived emotional-expression scores (S. Li & Deng, 2022; Meng et al., 2017). In a facial-expression recognition task, a model maps each image or segment to category probabilities or continuous scores for labels such as happiness, anger, sadness, fear, surprise, disgust, or neutral expression.

These score streams are valuable because many behavioral questions concern how visible affect unfolds over time. Researchers may ask whether expressive states persist or switch, whether they return after perturbation, and whether many model-output channels reflect a smaller number of expressive dimensions. Such questions arise in psychotherapy, autism assessment, collaborative work, public communication, and social-media video (Halamová et al., 2023; Holiday et al., 2023; Q. Li et al., 2024; Manelis-Baram et al., 2025).

Automated facial-analysis systems now make this kind of work feasible at scale. Datasets such as FER2013 supported deep learning approaches to facial-expression recognition, especially Convolutional Neural Networks (Arriaga et al., 2017; Ezerceli & Eskil, 2022; CNNs; Goodfellow et al., 2013). Most early systems inherited their emotion categories from the training data and focused on fixed classification tasks, commonly returning probabilities for seven categories: anger, disgust, fear, happiness, sadness, surprise, and neutral facial expression (Wang et al., 2022). More recently, CLIP-based and related vision-language models have enabled zero-shot classification with user-specified emotion labels (Foteinopoulou & Patras, 2023; H. Li et al., 2023; Radford et al., 2021; Tomašević et al., 2026), making it possible to represent each video frame as a vector of prompt-defined classifier scores.

The main obstacle is measurement. Emotional-expression scores are trained-model outputs that index visible expressive behavior through a scoring procedure. Facial-expression technologies can be interpreted as tools for emotion recognition or social-interaction analysis, and those interpretations answer different questions (Allison & McKeown, 2023). A separate psychometric question is whether multiple score channels should be treated as variables in their own right or as fallible indicators of lower-dimensional expressive processes (Mehu & Scherer, 2015; Nesselroade et al., 2013; Russell, 2003). Measurement error can alter the structure recovered by dynamic-network methods, so method validation must separate latent temporal organization from the observation process used to measure it (Kankaanpää et al., 2026).

The remaining methodological gap is a controlled generator that jointly manipulates latent expressive dynamics and a multichannel observation process resembling video-derived scores. Oscillator models provide

a language for level, rate of change, damping, and perturbation in intraindividual dynamics (Boker, 2015; Boker & Nesselroade, 2002; Ollero et al., 2025). Network and dimensional-recovery methods can analyze observed intensive series. Researchers also need a simulation framework in which the generating dimensionality, indicator blocks, measurement quality, and dynamic regime are prespecified before recovery methods are interpreted.

This paper introduces the Latent Oscillator Measurement Model (LOMM), a generative simulation model for model-derived emotional-expression score channels. LOMM first generates latent expressive trajectories from second-order oscillator dynamics and then maps those latent processes into classifier-output score channels through an explicit measurement layer. The model lets researchers control latent dimensionality, feedback regime, loading strength, measurement error, and activity pattern before testing whether dynamic methods recover the generating structure.

This study addresses four questions. First, can a calibrated LOMM reproduce selected distributional, temporal, spectral, and multivariate features of empirical video-derived emotional-expression scores? Second, across a broad factorial grid, how accurately and efficiently does DynEGA recover the generating dimensionality and indicator blocks? Third, on the same reduced set of generated panels, how do DynEGA, GraphicalVAR, and GIMME compare when their fitted structures are partitioned with the same community-detection algorithms? Fourth, how do activity regime, loading strength, measurement error, and generating dimensionality alter structural recovery, trajectory-score recovery, and computational feasibility?

These questions require two complementary simulation designs. The empirical realism check uses a softmax-linked bounded score panel so simulated scores can be compared with probability-like empirical classifier outputs. The dimensional-recovery simulations use pre-link linear block-measurement panels so that the recovery target remains the known measurement structure generated by LOMM. A broad DynEGA grid evaluates the focal dimensional-recovery method across the full factorial space, including two-, three-, and four-dimensional generating structures. A matched three-method grid uses a reduced design, including four indicators per latent dimension and 100 time points, so that DynEGA, GraphicalVAR, and GIMME can be applied to the same generated panels with the same downstream community-detection procedures. This split preserves a broad scalability assessment for DynEGA while providing a panel-by-panel comparison of three network-based or network-convertible recovery pipelines. Figure 1 links the empirical calibration, scenario construction, and recovery analyses used to answer these questions.

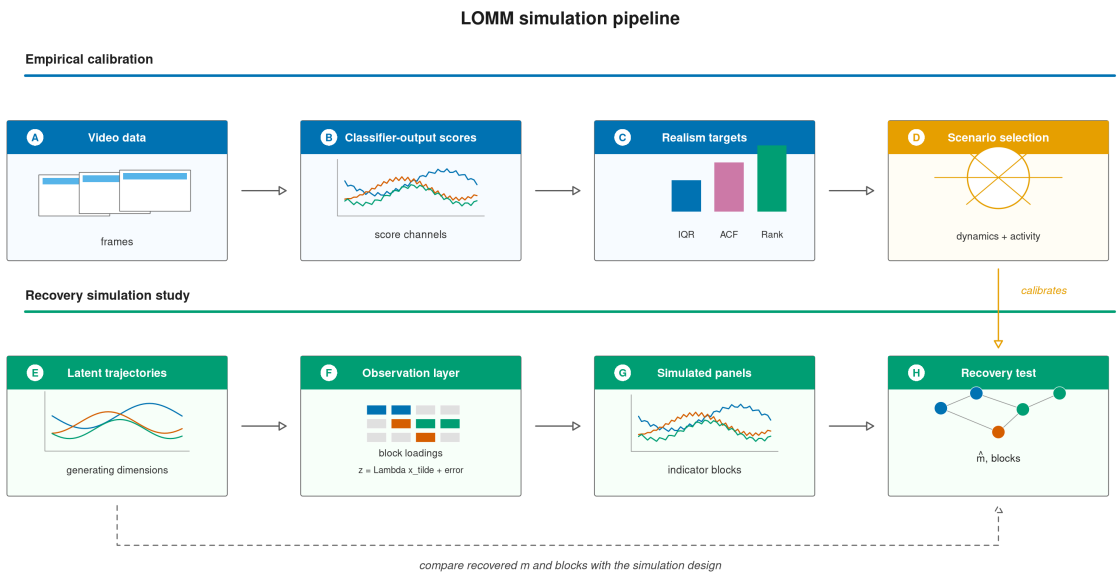


Figure 1: Visual overview of the LOMM simulation pipeline. Panels A–C show the empirical calibration step and realism targets for model-derived emotional-expression score series. Panel D shows scenario selection through dynamics, activity, and measurement parameters. Panels E–H show the recovery simulation from latent trajectories, through the observation layer and simulated score panels, to the recovery test. The recovery test compares recovered dimensions and indicator blocks with the simulation design.

The Latent Oscillator Measurement Model

LOMM starts from a measurement object: a video produces frame-level classifier-output score channels for visible expression. The simulation treats those channels as fallible indicators of lower-dimensional expressive processes. A short clip can contain low expressivity, brief expressive episodes, gradual recovery, persistent activation, or alternation between expressive regimes. The latent representation therefore needs to describe level, rate of change, return toward equilibrium, overshoot, damping, and perturbation, concepts that are central to affective-dynamics research (Hollenstein, 2007; Houben et al., 2015; Kuppens & Verduyn, 2017).

Oscillator models provide a compact language for those features. Prior work has used second-order oscillator representations to describe intraindividual dynamics in which a current level and its rate of change jointly determine subsequent movement (Boker, 2015; Boker & Nesselroade, 2002; Ollero et al., 2025). LOMM uses the same state representation for multivariate simulation and measurement evaluation. It combines three layers: a latent oscillator layer that generates expressive trajectories, an activity layer that controls which latent dimension is expressed over time, and a block measurement layer adapted from direct autoregressive factor-score modeling (H. Golino et al., 2022; Nesselroade et al., 2013).

The latent layer treats each expressive dimension as a second-order state process advanced in discrete simulation steps. For state $\mathbf{s}_{if,t} = [x_{if,t}, v_{if,t}]^\top$, the implemented exact linear step is

$$\mathbf{s}_{if,t+dt} = \Phi_f(dt)\mathbf{s}_{if,t} + \Gamma_f(dt)q_{if,t}, \qquad q_{if,t} \sim \mathcal{N}(0, \sigma_{qf}^2), \tag{1}$$

where

$$\Phi_f(dt) = \exp(\mathbf{A}_f dt), \qquad \mathbf{A}_f = \begin{bmatrix} 0 & 1 \\ \eta_f & \zeta_f \end{bmatrix} \qquad \text{and} \qquad \Gamma_f(dt) = \int_0^{dt} \exp(\mathbf{A}_f u)\mathbf{b}\, du, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

In Equation 1, i indexes videos, f indexes latent expressive dimensions, and t indexes discrete simulation time points spaced dt apart. The state variable $x_{if,t}$ is the latent level of dimension f for video i, and $v_{if,t}$ is its rate of change. The parameter η_f is position feedback, ζ_f is velocity feedback, $q_{if,t}$ is the stepwise Gaussian acceleration input used at each transition, and σ_{qf} is the standard deviation of that input. Within a step, the recursion is the exact transition of the linear second-order oscillator $\ddot{x}_{if} = \eta_f x_{if} + \zeta_f \dot{x}_{if} + q_{if,t}$ with the shock held constant, so $\Phi_f(dt)$ propagates the deterministic state and $\Gamma_f(dt)$ is the zero-order-hold loading through which $q_{if,t}$ enters as a per-step acceleration input.

The dynamic parameters set the latent regime. Negative η_f pulls a displaced latent trajectory back toward its equilibrium level, with larger negative values producing stronger restoring acceleration. Negative ζ_f attenuates velocity over time, $\zeta_f = 0$ removes velocity feedback, and positive ζ_f can sustain or amplify ongoing change over the finite interval being simulated. The reported simulations use an uncoupled multivariate system: η_f, ζ_f, and σ_{qf} may vary by factor, and off-diagonal dynamic effects are omitted. The state is initialized with $x_{if,1} = 0.25$ and $v_{if,1} = 2$.

The activity layer turns raw oscillator states into scenario-specific expression episodes. The trajectory passed to the measurement model is $\tilde{x}_{if,t} = a_{if,t}x_{if,t}$, where $a_{if,t} \geq 0$ is a scenario-defined activity envelope generated from active gains, background gains, and dwell schedules. In single-regime scenarios, one dimension has higher activity across the full sequence; in switching scenarios, the active dimension changes after sampled dwell intervals; and in low-expression scenarios, all dimensions remain close to background activity. The activity envelope changes dominance, switching, and expression strength while preserving the prespecified generating dimensions and indicator memberships.

The measurement layer maps the activity-scaled latent trajectories into simulated score channels. For video i, if each latent dimension has k observed indicators, then the number of observed score channels is $p = mk$ and the linear score vector is

$$\mathbf{z}_{it} = \Lambda_i \, \tilde{\mathbf{x}}_{it} + \varepsilon_{it}, \qquad \varepsilon_{it} \sim \mathcal{N}(\mathbf{0}, \Sigma_i). \tag{2}$$

In Equation 2, $\mathbf{z}_{it}$ is the p-channel linear score vector, $\tilde{\mathbf{x}}_{it}$ is the m-dimensional activity-scaled latent vector, Λ_i is a $p \times m$ loading matrix, ε_{it} is measurement error, and Σ_i is the error covariance matrix. Each observed score belongs to one indicator block B_f and loads on exactly one latent dimension: for indicator j in block B_f, $(\Lambda_i)_{jf} = \lambda + u_{ij}$ with baseline loading λ and loading jitter $u_{ij} \sim \mathrm{Unif}(-0.15, 0.15)$; all off-block loadings are zero. The measurement-error covariance matrix is diagonal, $\Sigma_i = \mathrm{diag}(\sigma^2_{\varepsilon,i1}, \ldots, \sigma^2_{\varepsilon,ip})$, with $\sigma^2_{\varepsilon,ij} = s^2_\varepsilon + r_{ij}$ and item-level variance jitter $r_{ij} \sim \mathrm{Unif}(-0.01, 0.01)$. The simulation grid reports s_ε as the measurement-error standard deviation, and s^2_ε enters the covariance matrix before item-level jitter is added on the variance scale.

These observed channels are classifier-output measurements of visible expression. The block model tests whether recovery methods recover a prespecified observation process under controlled loading strength, measurement error, dynamic regime, and activity regime. It leaves semantic label validity and subjective emotional experience outside the evidential scope of the simulation.

For the empirical realism check, the linear score vector is transformed to a bounded observed scale with a row-wise softmax link. If $\mathbf{z}_{it}$ is the pre-link score vector from Equation 2, the retained realism layer uses

$$y_{itj} = \frac{\exp(z_{itj})}{\sum_{\ell=1}^{p} \exp(z_{it\ell})},$$

equivalent to a softmax temperature of 1.00. The empirical realism analysis uses the softmax-linked $\mathbf{y}_{it}$ panel to compare simulated scores with bounded probability-like empirical scores. The dimensional-recovery simulations use the linear $\mathbf{z}_{it}$ score panel. In both uses, generating block membership is defined by the nonzero entries of Λ_i. Neutral can appear as an empirical classifier-output channel, but it is not assigned a source-truth latent factor in the final benchmark unless a neutral block is specified by design.

As an illustrative substantive mapping, a three-factor LOMM could represent approach-affiliation, threat-defense, and orienting-novelty dimensions. Prompt-defined classifier-output channels related to happiness, amusement, trust, or smiling could form an approach-affiliation block. Channels related to anger, fear, disgust, contempt, or anxiety could form a threat-defense block. Channels related to surprise, anticipation, confusion, or attention could form an orienting-novelty block. These labels are examples for simulation design, not validated affect categories in the empirical MAFW panels. The example draws on affect taxonomies that distinguish broad dimensions, discrete expression families, and opponent tendencies (Plutchik, 2001;

Russell, 1980, 2003; Russell & Mehrabian, 1977), and it is consistent with facial-expression benchmarks that combine categorical expression labels with dimensional ratings (Kollias et al., 2023; Mehu & Scherer, 2015; Mollahosseini et al., 2019). Alternative affective taxonomies can be encoded by changing the latent block map while keeping the LOMM architecture fixed.

The same three-factor example is shown in Figure 2 to make the role of the latent shock term visible.

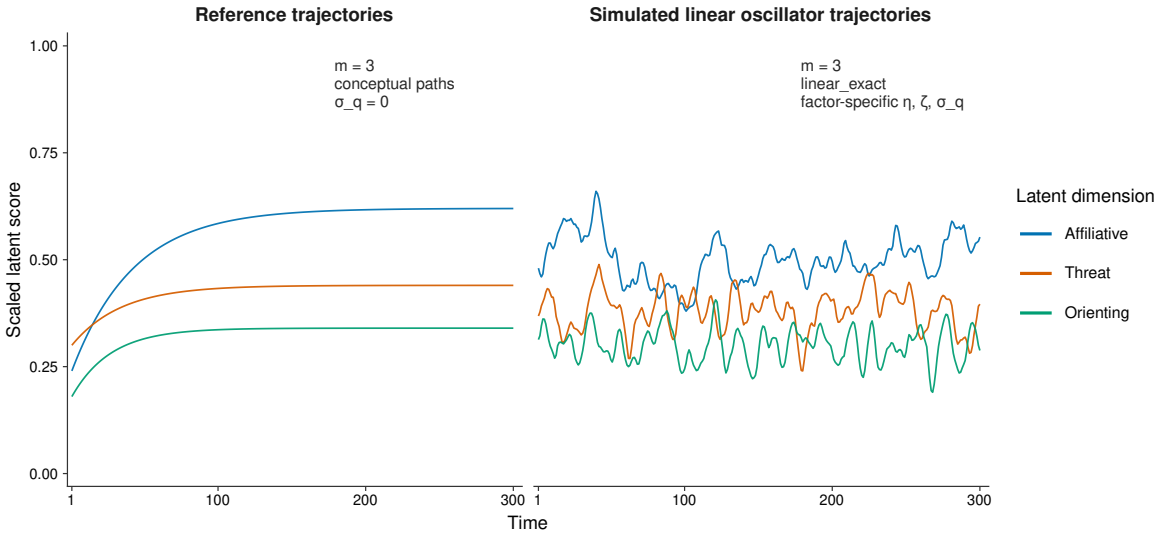


Figure 2: Three-factor LOMM example for theory-guided visible expressive dimensions. The left panel shows smooth reference trajectories without dynamic shocks; the right panel shows trajectories generated by the implemented linear oscillator process with factor-specific position feedback, velocity feedback, and innovation variance.

Figure 2 highlights the role of the innovation variance, σ_q^2, in LOMM. The reference panel shows the conceptual latent ordering without dynamic shocks: affiliative expression rises quickly and settles above the threat and orienting trajectories. The simulated oscillator panel retains this ordering while nonzero factor-specific σ_q perturbs each latent process at each time point, producing local peaks, reversals, and irregular fluctuations around the underlying trajectory. After approximately time point 100, affiliative expression (blue) remains the dominant latent tendency, while threat (orange) fluctuates at a moderate level and orienting (green) stays lower with shorter local excursions. Thus, σ_q turns a smooth conceptual model into latent score series that better resemble uneven temporal structure in model-derived visible-expression panels.

Method

The analyses follow the workflow in Figure 1. We first use empirical video-score panels to define selected realism targets for model-derived expression-score series (Panels A–C). We then translate those targets into

LOMM scenario conditions (Panel D) and evaluate dimensional-recovery methods on simulated panels whose dimensionality and indicator blocks are specified by the simulation design (Panels E–H).

Empirical Calibration and Realism Targets

Figure 3 illustrates the empirical score-series surface used to motivate the simulation study. MAFW is a large-scale, multimodal, compound affective database for dynamic facial-expression recognition in the wild (Liu et al., 2022). The dataset supplies naturalistic video clips for calibrating selected score-series statistics; it does not validate the classifier-output labels, the scoring model, or the subjective emotional states of the people in the clips. The realism check uses a 100-clip calibration sample scored with the same zero-shot procedure as the illustrated clip.

We used the transforEmotion R package to generate frame-level classifier-output scores from the calibration videos (Tomašević et al., 2026). The scoring call used model key `oai-large`, which maps in the local model registry to `openai/clip-vit-large-patch14`, corresponding to OpenAI CLIP ViT-Large/14. Each sampled video was decoded into frames and scored against the prompt-defined classifier-output channels labeled anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment. These 11 prompt channels follow the MAFW single-label schema and are model outputs rather than frame-level human annotations. Within each clip, short missing runs were interpolated, frames with complete usable scores were retained, and clipped score vectors were renormalized as probability-like outputs. These prompt-defined classifier-output panels provide empirical targets for the LOMM observation layer through their temporal persistence, variability, and co-fluctuation.

The selected-frame score profiles from the same MAFW clip anchor the empirical example (Panel A): different frames receive different values across channels labeled helplessness, fear, disgust, sadness, anxiety, and disappointment. The frame-level trajectories show that the highest mean channels change in bursts with irregular local movement (Panel B). The single-video DynEGA network partitions the same clip's non-neutral classifier-output trajectories (Panel C). This panel is an illustrative readout of model-output channels in one clip; the simulation study later tests recovery under a prespecified generating structure.

The empirical calibration sample was assembled at the clip level from a label-diverse MAFW manifest, with MAFW category labels used for sample diversity rather than validation targets. After interpolation of short missing runs and complete-row filtering, the retained panel contains 100 clips spanning the 11 MAFW single-label categories. Usable clip lengths ranged from 17 to 50 frames, with a median of 36.5 usable frames. These retained frame-score panels define the empirical distributions used for the realism targets, and the short clip

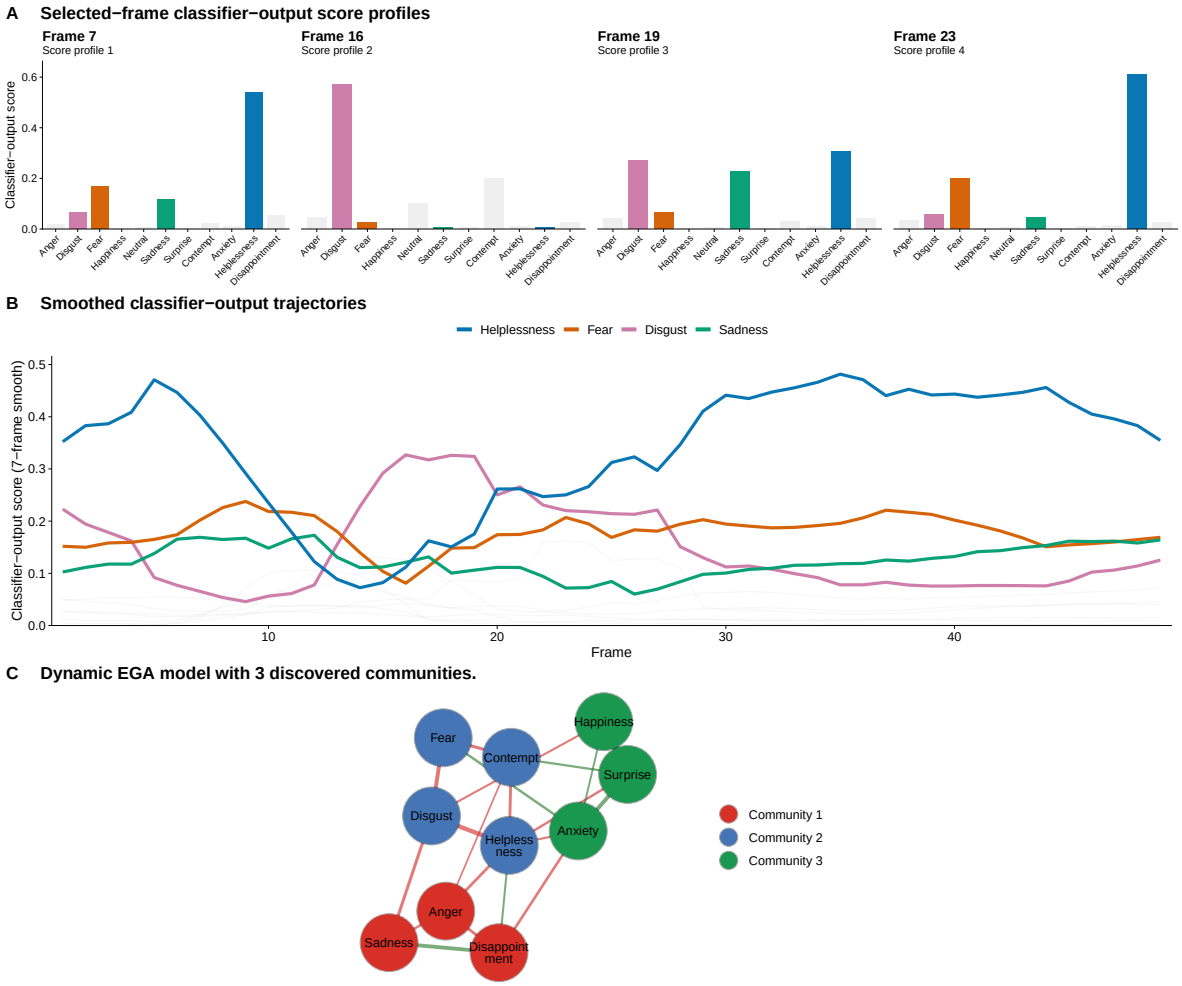


Figure 3: Single-video empirical score-series illustration from the MAFW dataset. Panel A shows selected-frame classifier-output score profiles from one clip; Panel B shows smoothed frame-level classifier-output trajectories for the highest mean channels; Panel C shows a DynEGA network estimated from the same clip's non-neutral classifier-output trajectories. Scores were generated with transforEmotion using model key `oai-large`, which maps to OpenAI CLIP ViT-Large/14 (`openai/clip-vit-large-patch14`).

lengths bound the interpretation of autocorrelation and spectral summaries.

For the empirical realism check, empirical and simulated panels are summarized at two levels. Observable-channel metrics are computed directly from the classifier-output score series within each clip. Recovered community-score summaries are computed after applying pooled DynEGA to the score panel and extracting score summaries for the recovered communities. This second readout is method-dependent descriptive information, not independent evidence that the empirical videos contain the same latent dimensions as the simulation. The observable-channel readout evaluates channel-level amplitude and movement; the recovered community-score readout evaluates the lower-dimensional geometry induced by this DynEGA summarization step.

The primary realism targets summarize dispersion, local movement, temporal structure, and effective multivariate dimensionality. Channel IQR is the interquartile range of a score series, so higher values mean wider score dispersion. Derivative IQR is the interquartile range of first differences, so higher values mean larger frame-to-frame movement. Lag-1 autocorrelation indexes local persistence. Run-length entropy is the entropy of monotonic run lengths in the first-difference sign series, so higher values indicate more varied directional regimes. Spectral entropy is the normalized entropy of the fast-Fourier-transform power distribution, so higher values indicate broader spectral complexity. Effective rank is the entropy-based rank of the clip-level correlation matrix, so higher values indicate more effective multivariate dimensionality in the score panel.

For channel IQR, derivative IQR, lag-1 autocorrelation, run-length entropy, and spectral entropy, the empirical and simulated distributions contain one value for each clip-by-channel series or clip-by-community-summary series. For effective rank, the distributions contain one value for each clip panel. For each metric distribution, we compare empirical and simulated values using a scaled median difference and a scaled one-dimensional Wasserstein distance. Both discrepancies are divided by the empirical interquartile range for that metric distribution, with a standard-deviation fallback when the empirical interquartile range is near zero. The larger discrepancy determines the classification: Pass for scaled discrepancy ≤ 0.33, Partial for scaled discrepancy ≤ 1.00, and Fail for scaled discrepancy > 1.00. Calls use the unrounded discrepancy values. These are construction thresholds, not validation thresholds; they do not validate the scoring model or the prompt labels.

Scenario Selection

Scenario selection links the empirical calibration targets to the simulation design. We group manipulated conditions into three parameter families (Panel D). The dynamics family varies the position feedback, velocity feedback, and innovation variance parameters, η_f, ζ_f, and σ_{qf}. The activity family varies whether one latent expressive dimension dominates across the full sequence, whether dominance switches after sampled dwell intervals, or whether all dimensions remain close to low-expression background activity. The measurement family varies the block loading matrix Λ_i and measurement-error covariance matrix Σ_i that generate the observed classifier-output score panels.

The scenario set contains six activity-and-dynamics families. Four families are generic score-panel regimes: low-expression single dominance, high-expression single dominance, high-expression switching, and high-expression rapid switching. Two families use the empirical calibration logic more directly: ranked-persistence and broad switching-geometry. Table 1 summarizes the activity envelope and dynamic rules used by each family. Across these families, LOMM varies the latent dimensionality and measurement layer so that the number of observed channels equals the number of latent dimensions times the number of indicators per dimension, $p = mk$. The broad DynEGA grid includes two-, three-, and four-dimensional generating structures, and the matched three-method grid uses two- and three-dimensional generating structures. The final crossed factors, replication counts, and method-specific grid sizes are specified in the dimensional-recovery simulation section below.

Table 1: Activity-and-dynamics scenario families used in the final simulations. Rows vary expression strength, switching, dwell length, feedback pattern, and innovation scale. Dwell values are fractions of the simulated sequence length.

Family	Source	Switching	Active/background gain	Dwell fraction	Dynamic rule
Low-expression single dominance	Generic score-panel regime	No	0.35 / 0.05	1.00 / 1.00	Template feedback; low innovation scale
High-expression single dominance	Generic score-panel regime	No	1.00 / 0.08	1.00 / 1.00	Template feedback; baseline innovation scale
High-expression switching	Generic score-panel regime	Yes	1.00 / 0.08	0.20 / 0.45	Template feedback; baseline innovation scale
High-expression rapid switching	Generic score-panel regime	Yes	1.25 / 0.08	0.03 / 0.10	Template feedback; high innovation scale
Ranked-persistence	Calibration-inspired regime	No	1.00 / 0.08	1.00 / 1.00	Ranked feedback; moderate innovation scale
Broad switching-geometry	Calibration-inspired regime	Yes	1.10 / 0.10	0.12 / 0.30	Ranked feedback; higher innovation scale

Numerical values for feedback and innovation-scale settings are reported in the supplementary scenario-

parameter tables.

For each simulated video, LOMM first generates latent trajectories from the scenario-specific dynamic parameters. The activity schedule then scales those trajectories: non-switching scenarios keep one active dimension for the full sequence, rotating the dominant dimension across videos, whereas switching scenarios sample dwell intervals and change the active dimension over time. The measurement layer is applied to the activity-scaled latent trajectories. Measurement error is reported as a standard deviation, with the corresponding variance entering the diagonal measurement-error covariance before small item-level variance jitter is added. The simulation design evaluates dimensional recovery under controlled combinations of dynamics, activity regimes, and measurement quality.

Dimensional-Recovery Simulation Study

The simulation begins with latent trajectories and the prespecified number of simulated expressive dimensions (Panel E). The LOMM observation layer maps each observed score channel to one latent dimension through the block loading matrix and adds item-level measurement error (Panel F), producing a pre-link simulated score panel (Panel G). The final step is the dimensional-recovery test, where the estimated number of dimensions and item memberships are compared against the generating measurement structure (Panel H).

The method comparison uses a common graph-partitioning target. Each pipeline receives the same simulated score panel, estimates a structure among observed score channels, converts that structure to communities, and compares the resulting partition with the generating indicator blocks. This comparison evaluates recovery of LOMM-generated score blocks, not the validity of classifier labels or method superiority for substantive emotion inference.

We compare Dynamic Exploratory Graph Analysis [DynEGA; H. Golino et al. (2022)] with GraphicalVAR (Epskamp et al., 2018; Epskamp, 2024) and GIMME (Gates & Molenaar, 2012; Lane et al., 2019) because they represent adjacent but distinct approaches to intensive longitudinal structure. DynEGA is the focal dimensional-recovery method: it estimates groups of variables whose changes are coordinated over time. In the present simulation, DynEGA tests direct recovery of the generating score blocks. GraphicalVAR is a dynamic-network comparator that separates lagged temporal relations from contemporaneous partial associations, and its fitted contemporaneous structure is partitioned after estimation. GIMME is an indirect feasibility-stress comparator: it estimates time-series paths for each clip treated as a separate time series, and the resulting contemporaneous structures are aggregated and partitioned. This use of GIMME treats clips as analytic series and should not be read as participant-level idiographic affect modeling.

DynEGA combines time-delay embedding and derivative estimation from dynamical systems with exploratory graph analysis for dimensionality assessment (Boker et al., 2010; H. Golino et al., 2020; H. F. Golino & Epskamp, 2017). We apply DynEGA to the stacked video panel with video identity supplied as the grouping variable. The embedding dimension is selected by minimizing the Total Entropy Fit Index (TEFI) over candidate values (H. Golino et al., 2021), using $n_{\text{embed}} \in \{5, 10, 15, \lfloor T/4 \rfloor, \lfloor T/2 \rfloor\}$ where admissible. DynEGA then estimates a regularized partial-correlation network (EBICglasso, the EGAnet default) from the pooled derivative panel, and that network is partitioned into communities, which are interpreted as recovered groups of score channels. The broad DynEGA grid uses Louvain community detection (Blondel et al., 2008) with the expand rule for unidimensional solutions. In the matched three-method grid, community detection is analyzed as a factor with Louvain, Leiden (Traag et al., 2019), and Walktrap (Pons & Latapy, 2005), holding the unidimensional rule at expand.

GraphicalVAR estimates a graphical vector autoregressive model for intensive longitudinal data (Epskamp et al., 2018; Epskamp, 2024). In this framework, the temporal network contains directed lag-1 regression parameters, whereas the contemporaneous network contains partial correlations among residuals at the same time point after lagged effects have been taken into account. In multilevel applications, the model can also distinguish within-person temporal and contemporaneous structure from between-person associations among stationary means. We fit GraphicalVAR to the stacked simulated score panel and use the estimated contemporaneous partial-correlation matrix as the structural matrix for dimensional partitioning, because LOMM recovery targets an undirected grouping of score channels. The diagonal is ignored and the resulting weighted matrix is treated as an undirected graph. Positive edge weights define the graph, with absolute weights used only when no positive edges are available. Louvain, Leiden, and Walktrap are applied to the same fitted contemporaneous structure and are analyzed as the community-detection factor.

GIMME is an automated time-series modeling approach built around unified structural equation modeling and vector autoregressive representations (Gates & Molenaar, 2012; Lane et al., 2019). The algorithm searches for lagged and contemporaneous paths that improve fit across the sample, retains paths that are common to many series, and then allows subgroup-shared or series-specific paths. In this simulation, each video clip is treated as the time series supplied to GIMME. We fit the VAR specification to the video-level score panels, extract the contemporaneous block from the clip-specific Ψ matrices, and average those blocks across clips to obtain a common structural matrix for dimensional partitioning. The diagonal is ignored and the averaged matrix is treated as an undirected graph. As with GraphicalVAR, positive edge weights define the graph, with absolute weights used only when no positive edges are available. Louvain, Leiden, and Walktrap are applied to the same fitted GIMME structure and are analyzed as the community-detection

factor.

All three fitted structures are converted to dimensional partitions and evaluated against the prespecified generating structure. DynEGA is evaluated on a broad grid for scalability and on the matched three-method grid; GraphicalVAR and GIMME are evaluated on the matched three-method grid because complete factorial estimation is computationally unstable in this setting. Community-detection algorithm is included as a diagnostic factor in the matched three-method grid; method and simulation design remain the primary organizing variables. Appendix A derives the population structural signal that each pipeline supplies to community detection under the linear block-measurement target and collects the pipeline differences in a single perturbation margin.

Performance is summarized using dimensional hit rate, assignment accuracy for observed score channels, matched trajectory-recovery correlation, matched trajectory-recovery normalized root mean squared error (NRMSE), runtime, and failure or timeout accounting. Method-readout rows are the denominator for dimensional hit rate and the status counts in Table 2; they include one row per method, generated panel, and community-detection readout. For GraphicalVAR and GIMME, feasibility is also reported as unique structure-fit status counts before the repeated Louvain, Leiden, and Walktrap readouts. Dimensional hit rate is the proportion of method-readout rows in which the method returns the correct number of latent dimensions. Assignment accuracy is computed for readouts with available partitions and asks whether observed score channels are assigned to the correct generating blocks after optimal matching of estimated communities to those blocks. For trajectory-recovery metrics, observed variables are standardized within the full panel and recovered community-score summaries are computed from the first principal component of each estimated block, with row means used when a principal-component score cannot be estimated. Matched trajectory-recovery correlation and NRMSE compare these recovered score summaries with the generating latent trajectories after sign-invariant matching between estimated communities and generating dimensions. These trajectory-recovery diagnostics are computed only when score summaries are available; they are secondary diagnostics rather than status-inclusive method-performance measures. Runtime is recorded for each model fit.

The final simulation uses a split design. The broad DynEGA grid varies the number of videos, series length, latent dimensionality, indicators per dimension, loading strength, and measurement-error standard deviation across 1,296 structural cells with 400 replications, yielding 518,400 DynEGA Louvain/expand method rows. The matched three-method grid uses four indicators per latent dimension, two- and three-dimensional generating structures, two sample sizes, 100 time points, three loading strengths, three measurement-error standard deviations, and all six scenario families. The reported final runs fix the innovation quality scale

at $Q = 0.75$, so the scenario base σ_q vectors listed in the supplementary scenario-parameter tables are the final-run innovation vectors. The latent systems were advanced with the exact linear transition of Equation 1 using step size $dt = 0.4$ for 100-point series and $dt = 0.25$ for 300-point series, so σ_{qf} is the per-step shock standard deviation at that resolution. This matched grid contains 216 structural cells with 100 replications. Louvain, Leiden, and Walktrap are included as community-detection levels, yielding 64,800 method-readout rows per method in the matched three-method grid. Timeouts, failed fits, and post-estimation extraction errors are retained in the reporting denominator because they are part of method behavior in applied simulation workflows.

Computational Environment

Final run metadata records R 4.5.1 on Rocky Linux 8.8 (`x86_64-redhat-linux-gnu`), 48 available logical cores, OpenBLAS 0.3.15, and LAPACK 3.9.0. Local inspection of the execution host for this final-results workspace reports dual Intel Xeon Silver 4310 CPUs, 48 logical threads, and 125 GiB RAM; the run artifacts store the operating system, R session, and available-core count but do not store RAM or a separate hostname. Key package versions were EGAnet 2.3.0, graphicalVAR 0.3.4, gimme 10.0, igraph 2.2.1, qgraph 1.9.8, psych 2.5.6, lavaan 0.6-20, dplyr 1.1.4, and ggplot2 4.0.1. Parallel jobs used one BLAS thread per worker. The broad DynEGA grid used 16 workers with one internal DynEGA core per fit, the matched DynEGA community-detection supplement used 32 workers, and the GraphicalVAR and GIMME matched-grid runs each used 24 workers with a 900-second elapsed-time limit per structure fit.

Runtime is elapsed wall-clock time, recorded with `proc.time()[["elapsed"]]` for successful fitted structures and summarized in seconds. For GraphicalVAR and GIMME, the recorded runtime includes the structure fit and post-fit extraction/community-detection readout; the same fitted structure is reused for Louvain, Leiden, and Walktrap, so feasibility accounting separates unique structure-fit status from repeated method-readout rows. Runtime summaries use successful fits, whereas timeout and error rows remain in feasibility and recovery denominators. The runtime results should therefore be read as within-machine feasibility estimates for this simulation workflow rather than hardware-independent speed benchmarks.

Results

Results are organized by the four research questions stated in the Introduction. Figure 4 gives the selected empirical-realism checks, Table 2 and Figure 5 summarize structural recovery, Figure 6 isolates simulated measurement-quality factors, and Figure 7 reports computational feasibility.

Empirical Realism Results

Before evaluating dimensional-recovery methods, we asked whether the calibrated LOMM realism panel reproduced selected distributional, temporal, spectral, and effective-dimensionality signatures of the MAFW-derived classifier-output score panels. Figure 4 summarizes this construction check at two readout levels: the observed score channels and the recovered community-score summaries extracted from the pooled DynEGA solution. The figure is generated from the empirical-realism file `metric_comparison_long.csv`, not from the final dimensional-recovery simulation outputs. This check evaluates selected score-series similarities for simulation construction; it does not assess general empirical realism, scoring-model validity, or emotion-label validity.

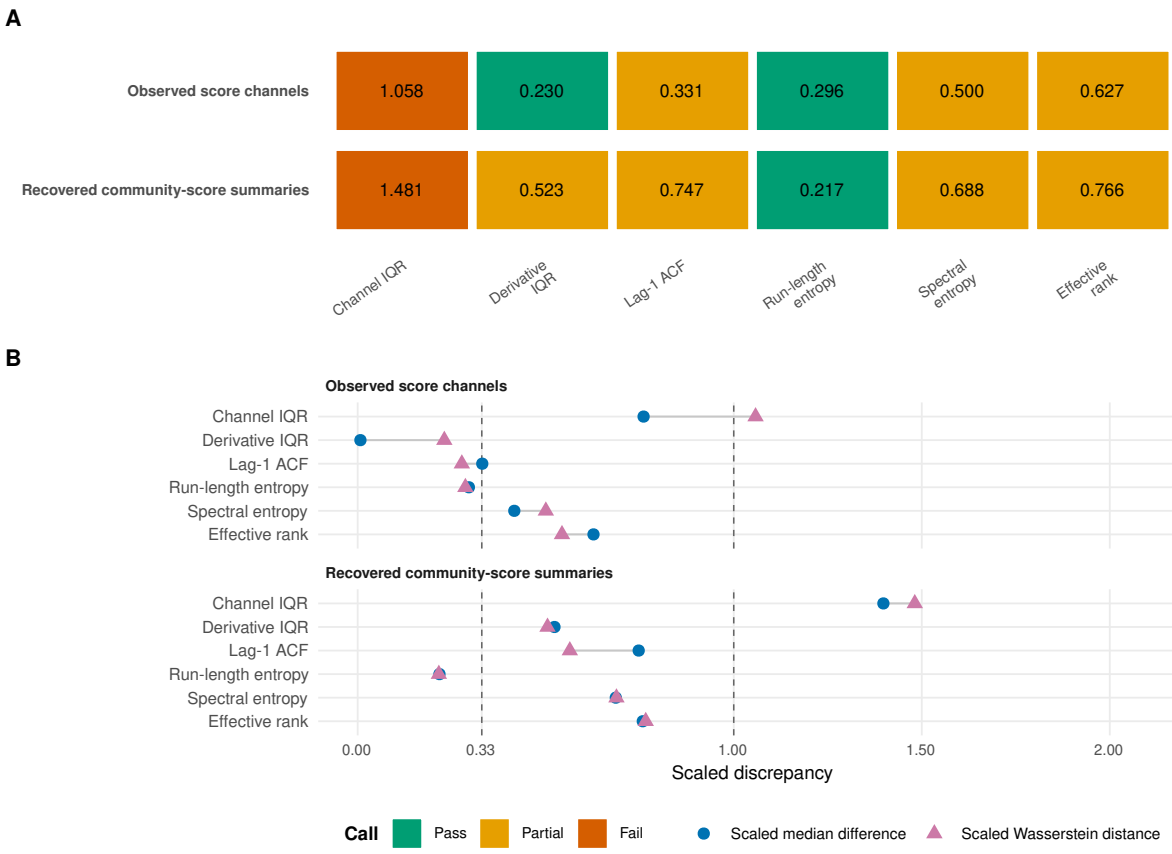


Figure 4: Selected empirical-realism checks for the calibrated LOMM realism panel. Panel A shows pass, partial, and fail classifications for six score-series targets at observed-channel and recovered community-summary readouts. Panel B shows the two discrepancy components separately; dashed reference lines mark the pass threshold of 0.33 and fail threshold of 1.00.

At the observed-channel level, the calibrated realism panel passed the derivative-IQR and run-length entropy checks. Lag-1 autocorrelation, spectral entropy, and effective rank were partial score-series matches because

they did not meet the strictest threshold. Channel IQR failed, showing that the simulated observed channels remained more dispersed than the empirical score channels. The softmax-linked panel is used for this realism check; the dimensional-recovery simulations use the linear block-measurement score panels described in the model section.

The recovered community-score-summary readout showed the same main boundary with a stronger amplitude mismatch. The community-score Channel IQR median was 0.832 in LOMM and 0.247 in the empirical panels, with a maximum scaled discrepancy of 1.481. This identifies amplitude dispersion as the main calibration boundary. Taken together, the selected realism checks support scenario construction for method testing; they do not establish empirical validity for the linear recovery panels.

The next analyses therefore evaluate known-truth recovery in the linear LOMM panels, not empirical emotion recognition or bounded-probability realism.

Simulation Results

The recovery analysis separates scalability, direct method comparison, simulated measurement-quality sensitivity, and computational feasibility. DynEGA was evaluated on the broad Louvain/expand grid and on the matched four-indicator design shared with GraphicalVAR and GIMME. The matched comparison aggregates across Louvain, Leiden, and Walktrap readouts and treats community detection as a diagnostic factor. Table 2 summarizes dimensional recovery, status counts, assignment accuracy, and successful-fit runtime by method.

Table 2: Overall method behavior in the final matched four-indicator grid, aggregated across Louvain, Leiden, and Walktrap readouts. Rows are method-community-detection readout rows. Hit rate is recovery of the generating number of latent dimensions using all readout rows as the denominator; Assignment is optimal block-assignment accuracy for readouts with available partitions. Median runtime is computed over successful fitted structures.

Method	Total rows	Valid rows	Timeouts	Errors	Hit (%)	Assignment (%)	Median runtime (s)
DynEGA	64800	64800	0	0	95.3	96.0	2.61
GraphicalVAR	64800	62616	2184	0	80.3	90.4	86.10
GIMME	64800	45882	12342	6576	25.9	67.2	280.50

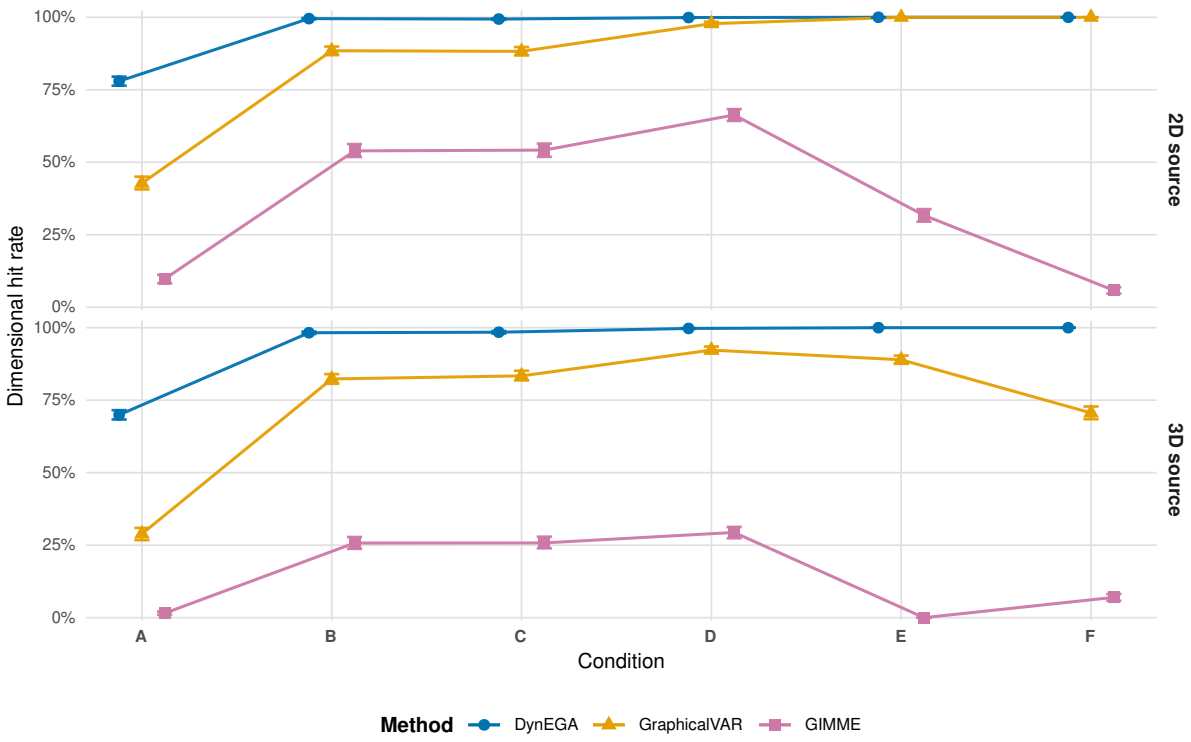


Figure 5: Dimensional hit rate across the final matched four-indicator method-comparison conditions. Lines show methods and vertical intervals show cluster-bootstrap 95% confidence intervals, resampling generated panels while retaining their community-detection readouts. Condition codes A–F refer to the simulated score-panel regimes in Table 1. Hit rates are status-inclusive and aggregated across Louvain, Leiden, and Walktrap readouts.

DynEGA was the only method evaluated on the broad grid and completed all 518,400 Louvain/expand rows. On that grid, it recovered the generating number of dimensions in 93.7% of rows and assigned observed classifier-output channels to the generating blocks with 95.2% accuracy. Table 2 reports status-inclusive method-readout rows after community-detection readouts. Unique structure-fit feasibility counts are reported in the Computational Feasibility section and Appendix D. Aggregating across Louvain, Leiden, and Walktrap, DynEGA recovered the generating number of dimensions in 95.3% of matched-grid rows, GraphicalVAR in 80.3%, and GIMME in 25.9%. Assignment accuracy followed the same ordering among readouts with available partitions: 96.0% for DynEGA, 90.4% for GraphicalVAR, and 67.2% for GIMME.

Figure 5 shows the matched-grid scenario pattern. DynEGA retained the strongest recovery profile across the simulated score-panel regimes. GraphicalVAR recovered the generating dimensionality in many ranked-persistence and broad-switching-geometry conditions when the structure fit completed, while recovery dropped in the low-expression single-dominance condition. GIMME showed the lowest status-inclusive recovery because many fits timed out or produced no usable contemporaneous structure for partitioning.

Because method and generated scenario family dominated dimensional hit rate, Figure 6 isolates the simulated measurement-quality factors that directly alter the observed score panels. The descriptive row-level factor screen showed the same ordering: method and scenario family had the largest associations with dimensional hit rate, measurement-error standard deviation also mattered, loading strength had a smaller association, and community detection contributed essentially no variation. These effect-size summaries are descriptive because community-detection readouts from the same fitted structure are repeated in the row-level data.

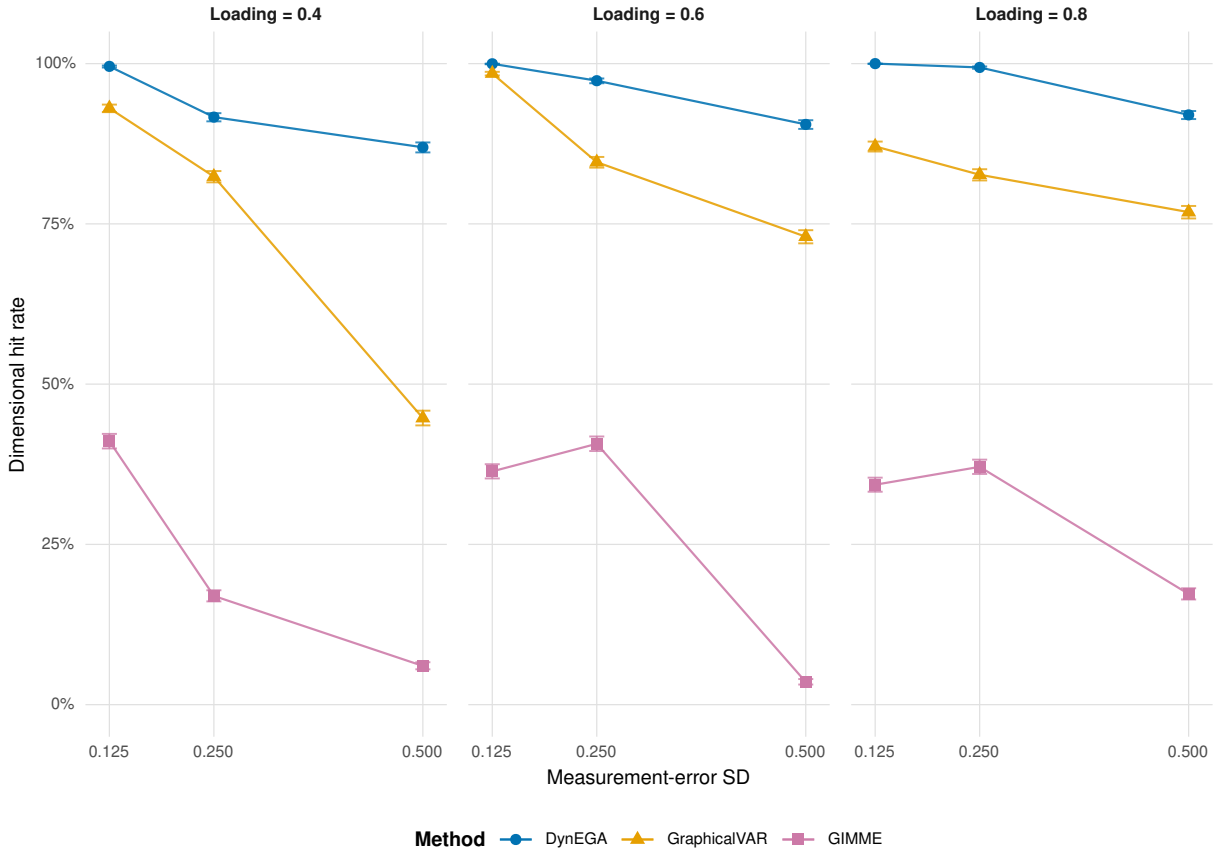


Figure 6: Dimensional hit rate stratified by measurement-error standard deviation and loading strength in the final matched four-indicator grid. Columns show loading strength; lines and symbols show methods. Hit rates are status-inclusive. Error bars are descriptive Wilson intervals computed from method-readout rows in each stratum, aggregated across community-detection readouts.

Figure 6 shows how method differences changed across measurement quality. DynEGA was most stable across the loading-by-error grid, although low-signal scenarios still reduced recovery. GraphicalVAR was more sensitive to the measurement layer: its hit rate was weakest when loadings were low and measurement error was high, and improved when observed channels carried a stronger signal from their generating dimensions.

GIMME remained lower on status-inclusive hit rate across most strata because errors and timeouts occurred before dimensional recovery could be evaluated. This loading-by-error gradient tracks the population signal-to-noise ratio $\lambda^2 V/\psi$ that drives both graph readouts in Appendix A.

Conditional on obtaining a usable structural readout, trajectory-recovery diagnostics followed the same measurement pattern. Recovery was strongest in the ranked-persistence and broad switching-geometry generated regimes and deteriorated as measurement-error standard deviation increased. These diagnostics are conditional summaries, not status-inclusive method-performance measures; full marginal means are reported in Appendix C.

Computational Feasibility

Computational feasibility was evaluated separately from recovery accuracy because the methods differed in both recovery behavior and scalability across the simulation design. Figure 7 summarizes successful-fit runtime in the matched four-indicator grid, and the text below accounts for broad-grid completion, timeouts, and post-estimation errors. This distinction is important for interpreting the comparator results: GraphicalVAR and GIMME are informative matched-grid comparisons with more limited scalability across the full LOMM grid.

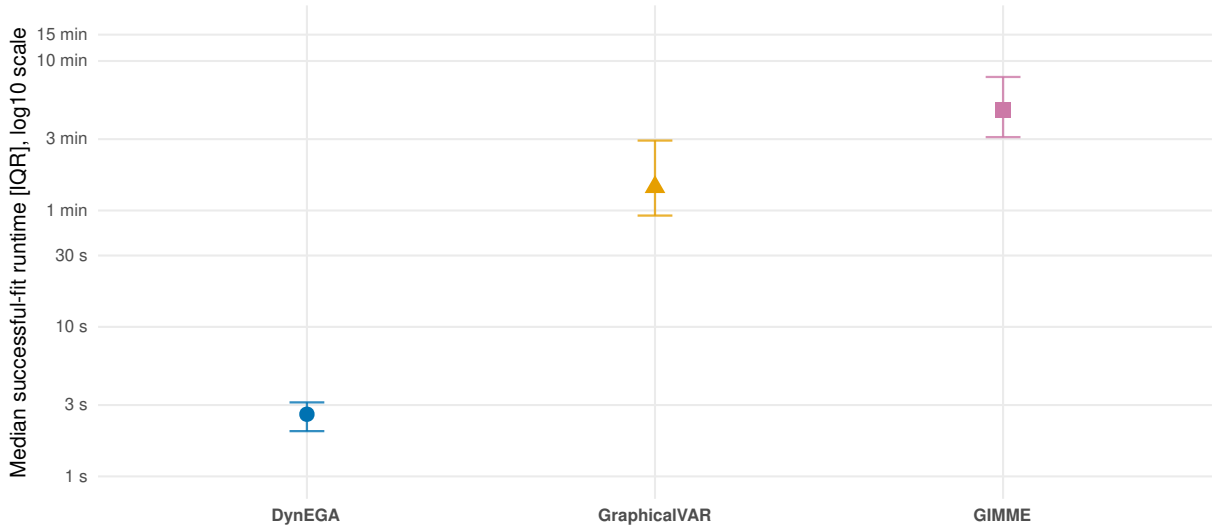


Figure 7: Runtime of the compared methods in the final matched four-indicator grid. The y-axis is shown on a log10 time scale. Points show median runtime for successful fitted structures and intervals show the interquartile range. For GraphicalVAR and GIMME, the same fitted structure is reused across Louvain, Leiden, and Walktrap readouts.

DynEGA was substantially faster than the two model-based comparators in the matched design. Median

successful-fit runtime was 2.61 s for DynEGA, compared with 86.1 s for GraphicalVAR and 280.5 s for GIMME. The broad DynEGA Louvain/expand grid also completed all 518,400 method rows without errors or timeouts. GraphicalVAR completed 20,872 of 21,600 matched-grid structure fits and timed out on 728 fits. GIMME completed 15,294 of 21,600 matched-grid structure fits, timed out on 4,114 fits, and returned post-estimation errors for 2,192 fits. Because GraphicalVAR and GIMME apply three community-detection algorithms to each fitted structure, these feasibility counts refer to unique structure fits before repeated CD readouts; Appendix D reports the fit-level and readout-row accounting side by side. In these LOMM simulation designs, DynEGA combined the strongest structural recovery with broad-grid scalability, GraphicalVAR was informative but slower and more condition-sensitive, and GIMME was feasibility-limited.

Discussion

This study introduced LOMM as a controlled generator of multivariate, video-derived emotional-expression score panels and used it to evaluate dynamic dimensional recovery. The findings are organized by the same four questions that guided the analysis. The selected empirical-realism checks reproduced local movement and directional-change texture, partially matched persistence, spectral complexity, and effective multivariate dimensionality, and retained a clear amplitude-dispersion mismatch. DynEGA recovered the generating dimensionality and indicator blocks most consistently in these LOMM simulation designs, across the broad focal-method grid and the matched three-method comparison. Recovery weakened when simulated measurement signal was lower, while GraphicalVAR was informative but condition-sensitive and GIMME was feasibility-limited in the matched design. The trajectory-recovery diagnostics add a conditional result among method-readout rows with recovered score summaries rather than a status-inclusive method ranking.

Appendix A frames recovery in the linear panels as a method-specific graph perturbation problem. Under the linear LOMM measurement model, indicators assigned to the same latent oscillator share a common change signal, whereas indicators assigned to different oscillators are independent. The measurement signal within each dimension is rank-one, and per-channel filtered population readouts form disconnected weighted cliques under the symmetric block target. DynEGA reads this structure through smoothed derivative estimates, whereas GraphicalVAR reaches the same block algebra through a multivariate VAR innovation projection under the simplified population target. GIMME adds an indirect graph-conversion and completion layer rather than a closed-form population block edge. The population edge sets how much common block signal reaches community detection, and recovery becomes more difficult when the method-specific perturbation approaches the weakest within-block edge.

This population structure is closely aligned with DynEGA's target in these LOMM simulation designs. DynEGA groups variables whose estimated changes are coordinated over time, so the common derivative signal within each LOMM block provides direct evidence for community recovery. Its high dimensional hit rate, high block-assignment accuracy, and complete broad-grid coverage indicate that this signal usually remained identifiable across the simulated score-panel regimes. The reductions under weak loadings, high measurement error, and low-expression generated regimes identify conditions in which the observed channels no longer carry enough common dynamic variation for reliable partitioning.

Beyond the method comparison, LOMM provides a reusable way to generate multivariate behavioral time series in which latent dynamics, activity regimes, and measurement quality can be manipulated separately. The oscillator layer controls restoring forces, momentum, and stochastic perturbation. The activity layer controls dominance and switching. The measurement layer controls indicator strength and error before scores are placed on their observed scale for empirical-realism checks. This separation supports targeted tests of whether an analysis recovers latent organization or responds primarily to features of the observation system.

GraphicalVAR provides a less direct representation of the generating measurement blocks. In this comparison, community detection is applied to the contemporaneous partial-correlation matrix, which conditions each pairwise association on the remaining channels. Appendix A shows that, in the symmetric block model, the population VAR innovation readout preserves block support while compressing the within-block edge toward a ceiling set by block size, and that the VAR projection removes the predictable part of the common block signal before that readout. The observed sensitivity of GraphicalVAR is consistent with this mechanism, especially when loadings are weak or measurement error is high, but the simulation does not isolate this population attenuation from lag estimation, regularization, finite-sample variation, or timeout behavior. In the matched grid, GraphicalVAR completed most structure fits and had 728 unique structure-fit timeouts, so its main limitation in this comparison was condition sensitivity with some timeout loss.

GIMME has broader native aims than dimensional partitioning. It searches for group, subgroup, and individual lagged and contemporaneous paths, whereas the present evaluation treats video clips as analytic time series, averages clip-specific contemporaneous structures, and applies a separate community-detection step. Its lower status-inclusive hit rate therefore reflects both this indirect GIMME-to-community readout and substantial feasibility costs, including long successful-fit runtimes, timeouts, and post-estimation errors. The result evaluates the implemented clip-level GIMME-to-community pipeline under this design. Person-level idiographic applications of GIMME require separate evidence.

Conditional on a valid structural readout, trajectory-recovery diagnostics showed the largest marginal dif-

ferences by generated scenario family and measurement-error standard deviation. Ranked-persistence and broad switching-geometry panels produced the closest trajectory summaries, whereas low-expression single-dominance panels were the most difficult. These diagnostics evaluate recovery of simulated latent trajectories after a usable structure has been estimated. They do not evaluate classifier-output validity, emotion-label validity, or status-inclusive method performance. Full marginal summaries are reported in Appendix C.

The empirical realism check sets the boundary on this interpretation. It evaluates selected statistical resemblance between simulated panels and classifier-output score panels from the MAFW calibration sample; it does not validate the scoring model, the prompt labels, or subjective emotional states in the videos. The calibration used one dataset label schema and one scoring implementation, and the empirical clips were short relative to some simulation regimes. The short empirical clips support calibration of selected local score-series properties, whereas the longer simulation conditions test method behavior under controlled series lengths. The current softmax-linked realism panel reproduces several local temporal and effective-dimensionality features, but the amplitude-dispersion failure shows that the observation layer still leaves marginal score dispersion too high.

Several limitations follow directly from this design. The method-comparison results apply to the linear block-measurement recovery panels, while the softmax-linked panels were used for empirical-realism calibration. The present LOMM implementation uses independent oscillator dimensions, activity-gated block loadings, and diagonal measurement error. These choices make the recovery target interpretable, but they simplify the geometry of visible expressive behavior and the dependence structure of classifier-output score channels. The method comparison covers network-based or network-convertible recovery pipelines, not all dynamic latent-variable approaches. Future work should test richer oscillator coupling, cross-loadings, correlated measurement error, alternative bounded or compositional observation layers, additional scoring systems, and empirical applications where simulation results guide method validation before substantive interpretation of classifier-output score dynamics. The present results support LOMM as a controlled validation environment for selected temporal and dimensional properties. Full generative theories of affect and validation of scoring-model emotion labels require separate evidence.

Declarations

Funding

The authors received no specific funding for this work.

Competing interests

The authors declare that they have no competing interests.

Ethics approval

This study used simulations and secondary, publicly described reference materials. It did not involve new recruitment, intervention, or direct collection of human-participant data by the authors.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The manuscript analyses use simulation outputs and derived classifier-output score summaries generated for this study. The OSF archive ([anonymized OSF link provided for masked review]) contains the version-controlled manuscript repository, final simulation runners, figure-generation scripts, manuscript-facing derived outputs, provenance manifests, and from-scratch rerun instructions for the final simulation grids. Raw MAFW video materials are third-party dataset materials; access to the original videos is governed by the MAFW dataset providers. Derived empirical score summaries needed for the realism check are included in the archive.

Code availability

All code needed to rerun the final LOMM simulations, rebuild the method-comparison summaries, regenerate the manuscript figures and tables, and render the manuscript is available in the archived project repository ([anonymized OSF link provided for masked review]). The repository includes the R simulation

runners, empirical realism and method-comparison scripts, manuscript rendering source, claim ledger, and run manifests.

Authors' contributions

Aleksandar Tomasevic contributed conceptualization, formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Hudson Golino contributed formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Alexander P. Christensen contributed methodology, software, writing the original draft, and reviewing and editing the manuscript.

Open Practices Statement

The study was not preregistered. The OSF archive ([anonymized OSF link provided for masked review]) contains analysis code, simulation code, manuscript source files, generated figure and table inputs, derived empirical score summaries needed for the realism check, and provenance materials for rerunning the final simulations from scratch. Raw MAFW video data are third-party materials governed by the dataset providers.

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Appendix A: Perturbation Targets for LOMM Recovery

This appendix gives a population argument for the method-comparison results. LOMM recovery can be viewed as a graph perturbation problem. Each pipeline produces a population structural signal for the same block-measured linear score panel, and the fitted readout adds perturbations from finite samples, estimation choices, convergence behavior, extraction, and community detection.

The argument reaches three conclusions. For the idealized DynEGA and GraphicalVAR population readouts, the generating block support is preserved exactly under the symmetric linear block target. Their within-block partial-correlation edges share one ceiling and differ through the signal-to-noise ratio delivered by the derivative or VAR innovation readout. GIMME has a different mathematical status and is treated separately as an indirect graph-conversion pipeline whose recovery also depends on path search, convergence, extraction, averaging, and completion. Section A.1 defines the symmetric block target, A.2 derives the two graph readouts available for per-channel filtered panels, A.3 maps DynEGA and GraphicalVAR to their population readouts, A.4 gives the shared perturbation margin, A.5 treats GIMME's indirect readout and completion, and A.6 summarizes the comparison.

A.1 Linear Block Target

Consider a population LOMM with m latent dimensions, k observed indicators per dimension, and $p = mk$ observed score channels. Let $B_1, \dots, B_m$ denote the true indicator blocks. For indicator $j \in B_f$, use the equal-loading, equal-error measurement model

$$Y_j(t) = \lambda Z_f(t) + \varepsilon_j(t), \qquad Z_f(t) = a_f(t) X_f(t),$$

where $Z_f(t)$ is the activity-scaled latent oscillator for dimension f, $\lambda > 0$, and $\varepsilon_j(t)$ is measurement error with variance $\psi = s_\varepsilon^2$. Measurement errors are independent across indicators and time in this target, so $\mathrm{Cov}(\varepsilon_j(t), \varepsilon_\ell(s)) = 0$ whenever $j \neq \ell$ or $t \neq s$. The latent processes Z_f are zero-mean and mutually independent in this simplified population object. Activity gating makes Z_f nonstationary by design, so the variances used in this appendix are time-averaged second moments over the analyzed window. Loading jitter, item-level variance jitter, switching schedules, initialization transients, finite-sample activity imbalance, and estimator behavior are treated as perturbations around this symmetric target.

For a method readout r, write the fitted graph as

$$\widehat{\mathbf{W}}^{(r)} = \mathbf{W}_*^{(r)} + \mathbf{E}^{(r)}.$$

Here $\mathbf{W}_*^{(r)}$ is the population structural matrix induced by the method readout, and $\mathbf{E}^{(r)}$ collects finite-sample variation, derivative estimation, model estimation, regularization, convergence behavior, extraction errors, and community-detection effects. This notation keeps the comparison focused on two quantities: the population signal available to the method and the perturbation added by the implemented pipeline.

A.2 Filtered Panels and Two Graph Readouts

Two useful population graph readouts can be derived for any per-channel filtered block panel. DynEGA estimates local derivatives of every channel, and a lag difference is a simple analytic example of the same channelwise operation. GraphicalVAR is handled separately in A.3 because its contemporaneous innovations come from a multivariate VAR projection rather than from an identical filter applied to each channel.

Let L denote a linear filter applied to each channel separately. Applying L to the measurement model for $j \in B_f$ gives

$$(LY_j)(t) = \lambda\,(LZ_f)(t) + (L\varepsilon_j)(t),$$

so the filtered panel keeps the one-factor block structure. Write $V_{L,f} = \mathrm{Var}(LZ_f)$ for the filtered common-signal variance and $\psi_L = \mathrm{Var}(L\varepsilon_j)$ for the filtered noise variance. Because L acts on each channel separately, filtered errors stay independent across channels, and the within-block covariance matrix of the filtered channels is

$$\Sigma_{L,f} = \lambda^2 V_{L,f}\,\mathbf{1}_k\mathbf{1}_k^\top + \psi_L I_k, \qquad \theta_{L,f} = \frac{\lambda^2 V_{L,f}}{\psi_L},$$

with zero cross-block covariance. The filtered signal-to-noise ratio $\theta_{L,f}$ is the only quantity through which the filter enters the population graph. Filtered errors can be serially correlated; the population readouts below use only the contemporaneous covariance across channels, and serial dependence is absorbed into the estimation perturbation in A.4.

The filtered panel admits two graph readouts. The marginal readout uses correlations directly. For two distinct indicators $j, \ell \in B_f$,

$$w_f^M(L) = \left|\mathrm{Cor}(LY_j, LY_\ell)\right| = \frac{\theta_{L,f}}{1+\theta_{L,f}}.$$

The partial readout conditions each pair on the remaining channels. Using the Sherman-Morrison identity,

$$\Sigma_{L,f}^{-1} = \psi_L^{-1}\left[I_k - \frac{\theta_{L,f}}{1+k\theta_{L,f}}\mathbf{1}_k\mathbf{1}_k^\top\right],$$

and the absolute partial correlation between two indicators in the same block is

$$w_f^P(L) = \frac{\theta_{L,f}}{1+(k-1)\theta_{L,f}}.$$

Because the filtered precision matrix is block-diagonal, cross-block partial correlations are exactly zero. These marginal and partial readouts therefore preserve the generating block support without error for the per-channel filtered population object, and the population graph has block-clique form for any per-channel filter,

$$\mathbf{W}_* = \mathrm{blockdiag}\left(w_1(J_k - I_k), \ldots, w_m(J_k - I_k)\right), \qquad w_f \in \{w_f^M(L),\, w_f^P(L)\}.$$

The two readouts differ in how much within-block signal they keep. For the same signal-to-noise ratio θ,

$$\frac{w^P(\theta)}{w^M(\theta)} = \frac{1+\theta}{1+(k-1)\theta},$$

so $w^P(\theta) < w^M(\theta)$ whenever $k > 2$ and $\theta > 0$. As $\theta \to \infty$,

$$w^P(\theta) \to \frac{1}{k-1}, \qquad w^M(\theta) \to 1.$$

The partial readout compresses an arbitrarily strong common-source signal toward a ceiling set by block size alone: with four indicators per dimension, no within-block partial-correlation edge can exceed 1/3. Both readout maps are increasing in θ, so filter and readout separate cleanly. The filter sets $\theta_{L,f}$, and the readout maps $\theta_{L,f}$ to the edge weight supplied to community detection.

Figure A1 shows both readout maps for the matched grid's four indicators per dimension. Neither implemented pipeline uses the marginal readout; it is the pre-conditioning benchmark that shows how much

common-source signal exists before conditioning, and the vertical gap between the curves is the compression that the partial readout applies to that signal. The shaded band marks the $\lambda^2/s_\varepsilon^2$ span of the matched measurement grid, so the simulated loading and error conditions place $\theta_{L,f}$ inside the band up to the filtered common-signal variance $V_{L,f}$. Across that span, the marginal curve rises from mid-range values toward 1, whereas the partial curve is already near its 1/3 ceiling: strong measurement conditions widen the margin for the common-source signal but cannot widen the partial-correlation edge beyond the ceiling.

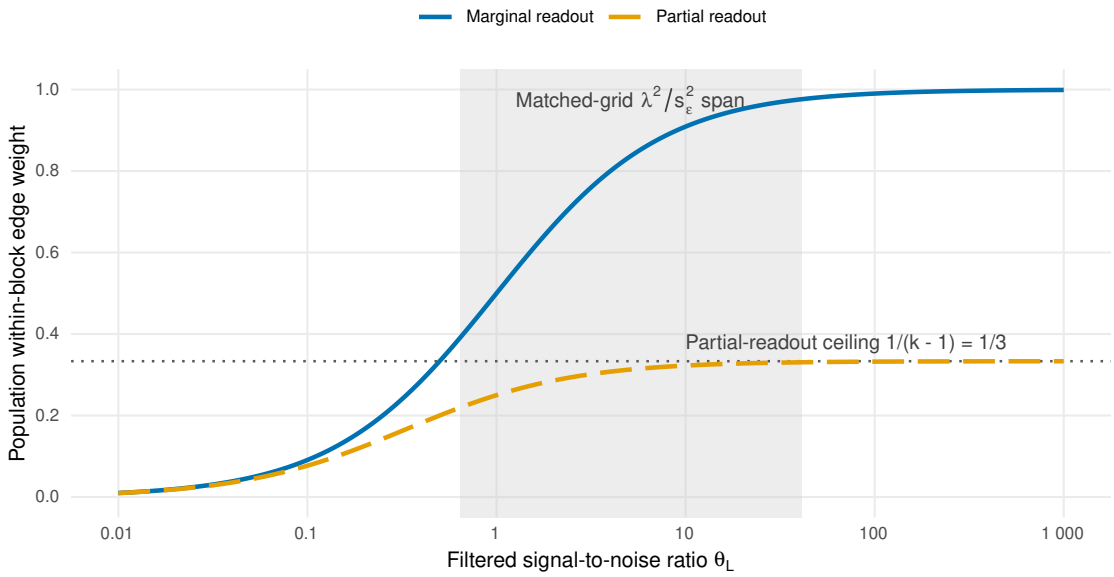


Figure A1: Population within-block edge weight as a function of the filtered signal-to-noise ratio for the marginal and partial readouts with four indicators per dimension. The dotted line marks the partial-readout ceiling $1/(k-1) = 1/3$. The shaded band spans the loading-to-error-variance ratios of the matched simulation grid; the filtered common-signal variance multiplies the signal-to-noise ratio within this band. Higher edge weight means more within-block signal margin before estimation perturbations.

A.3 Method-Specific Filters and Signal-to-Noise Ratios

The implemented pipelines are specific readout combinations. DynEGA estimates generalized local linear approximation (GLLA) derivatives of each score channel over a time-delay embedding window and then estimates a regularized partial-correlation network (EBICglasso) on the derivative panel. Its population object is therefore the partial readout of a per-channel derivative filter, $w_f^P(L_{\mathrm{GLLA}})$. A lag-h difference, $\Delta_h Y_j(t) = Y_j(t+h) - Y_j(t)$, gives a transparent analytic stand-in for that filter, with

$$\theta_f^{\Delta} = \frac{\lambda^2 \operatorname{Var}(\Delta_h Z_f)}{2\psi},$$

because independent measurement errors contribute 2ψ to the variance of every differenced channel. Both the lag difference and GLLA derivative estimation operate channel by channel. For the smooth oscillator trajectories generated here, the GLLA filter improves on this stand-in: it averages frame-level measurement noise over the embedding window while tracking the common latent change signal, which supports a high filtered signal-to-noise ratio.

GraphicalVAR reads partial correlations from the contemporaneous innovations of a lag-1 vector autoregression, so its population object is the partial readout of a multivariate VAR innovation projection, $w_f^P(L_{\mathrm{VAR}})$. For the population lag-1 fit of the symmetric block model, the implied coefficient matrix is proportional to $\mathbf{1}_k\mathbf{1}_k^\top$ within each block, and the innovation covariance retains the one-factor-plus-diagonal form. The compression map of A.2 therefore applies at the innovation scale under this symmetric target. The VAR projection also removes the predictable part of the common signal by construction: the simulated oscillator trajectories are strongly autocorrelated, so the lagged panel explains much of the common variance, while the serially independent measurement error passes into the innovations. The innovation readout therefore tends to enter the shared compression map with a smaller signal-to-noise ratio than a derivative filter applied to the same panel.

The population comparison between DynEGA and GraphicalVAR then has a precise form. Both population readouts supply partial-correlation edges to community detection, so both within-block edges are capped at $1/(k-1)$ and the generating block support is preserved in both population objects under the symmetric target. The population difference is where the derivative and innovation signal-to-noise ratios θ_{GLLA} and θ_{VAR} place each edge below that shared ceiling. The remaining differences are pipeline perturbations, which A.4 collects in a single margin.

A.4 Shared Perturbation Margin

Let a method's population graph have block-clique form

$$\mathbf{W}_*^{(r)} = \mathrm{blockdiag}\left(w_1^{(r)}(J_k - I_k), \ldots, w_m^{(r)}(J_k - I_k)\right),$$

where edge weights are on the nonnegative scale supplied to community detection and all m blocks share the same size k. For DynEGA and GraphicalVAR, $w_f^{(r)} = w_f^P(L_r)$ with the method-specific filter from A.3. Suppose every off-diagonal edge is estimated within δ_r of its readout-specific population value:

$$\max_{j \neq \ell} \left| \widehat{W}_{j\ell}^{(r)} - W_{*,j\ell}^{(r)} \right| \leq \delta_r.$$

For any node $j \in B_f$, the estimated total connection to its true block is at least

$$(k-1)(w_f^{(r)} - \delta_r),$$

whereas its estimated total connection to any wrong block is at most

$$k\delta_r.$$

Thus each node has stronger total connection to its true block than to any wrong block whenever

$$(k-1)(w_{\min}^{(r)} - \delta_r) > k\delta_r,$$

or equivalently

$$\delta_r < \frac{k-1}{2k-1} w_{\min}^{(r)}.$$

For the matched grid with $k = 4$, the sufficient node-level block-preference and threshold-separation condition is

$$\delta_r < \frac{3}{7} w_{\min}^{(r)}.$$

This margin is not a guarantee that Louvain, Leiden, or Walktrap recover the blocks, because exact recovery depends on the implemented community-detection algorithm. The bound has a narrower consequence. Because $\frac{k-1}{2k-1} < \frac{1}{2}$, the condition forces every estimated cross-block edge below every estimated within-block edge, so any threshold $\tau \in (\delta_r,\, w_{\min}^{(r)} - \delta_r)$ deletes all cross-block edges, keeps each block a connected clique, and reproduces the generating blocks exactly as the connected components of the thresholded graph.

The margin treats perturbation as symmetric noise on every edge, which makes δ_r a worst-case envelope. Regularized estimation acts asymmetrically: glasso shrinkage lowers within-block edges, which reduces $w_{\min}^{(r)}$, and sets weak cross-block edges to zero, which supports block separation.

The two population partial-readout pipelines share the same ceiling. DynEGA and GraphicalVAR supply partial-correlation edges to community detection, so $w_{\min}^{(r)} \leq 1/(k-1)$ for both pipelines, and with $k = 4$ the sufficient perturbation margin approaches at most

$$\delta_r < \frac{3}{7} \cdot \frac{1}{3} = \frac{1}{7}$$

in the high-signal limit. The population comparison therefore reduces to the two quantities identified in A.3: how close each filtered signal-to-noise ratio keeps $w_{\min}^{(r)}$ to the shared ceiling, and how large each pipeline's realized perturbation δ_r is. The observed recovery ordering is consistent with the derivative filter preserving more within-block signal than the innovation filter, while the full graphical vector-autoregression fit adds larger estimation and timeout perturbations.

The margin also identifies which blocks are fragile in each scenario family. Activity gating enters the filtered signal directly: $V_{L,f}$ is the time-averaged variance of the filtered activity-scaled process, so a dimension held at background gain contributes a small $\theta_{L,f}$ and a weak within-block edge. In the single-dominance families, background dimensions run at gains of 0.05–0.08 against active gains of 0.35–1.25 (Appendix E), so $w_{\min}^{(r)}$ is set by the background blocks, and in the low-expression family every dimension sits near background activity. The margin therefore predicts the scenario-family ordering observed in Figure 5 for DynEGA and GraphicalVAR: recovery is hardest where the activity schedule holds the weakest block's signal lowest. For GIMME, the same activity envelope is an expected stressor, but its effect is mediated by path search, convergence, extraction, averaging, graph conversion, and completion.

A.5 GIMME as an Indirect Graph-Conversion Readout

The implemented GIMME comparator has a different mathematical status. Its readout can be summarized as

$$\widehat{\mathbf{W}}^{G} = \Pi\left(\frac{1}{V}\sum_{i=1}^{V}\widehat{\Psi}_i\right),$$

where $\widehat{\Psi}_i$ is the clip-specific contemporaneous structure extracted from GIMME for clip i, V is the number of clips, and $\Pi(\cdot)$ denotes conversion to the undirected graph supplied to community detection. This readout depends on path search, clip-level model selection, convergence, extraction, averaging, and graph conversion. We therefore summarize GIMME as an indirect graph-conversion pipeline without a closed-form block edge w_f^G.

Status-inclusive recovery also includes completion. For any pipeline,

$$\Pr(\text{hit}) = \Pr(\text{usable graph}) \Pr(\text{hit} \mid \text{usable graph}).$$

The reported feasibility tables make this decomposition concrete. DynEGA returned usable graphs for every broad-grid and matched-grid row, so its status-inclusive and conditional hit rates coincide at 95.3%. GraphicalVAR completed 20,872 of 21,600 matched-grid structure fits (62,616 of 64,800 readout rows), so its 80.3% status-inclusive hit rate corresponds to a conditional hit rate of about 83% among usable readouts. GIMME completed 15,294 of 21,600 structure fits (45,882 of 64,800 readout rows), so its 25.9% status-inclusive hit rate corresponds to a conditional hit rate of about 37%. The decomposition separates GIMME's two losses: many fits return no usable graph, and partition quality after a usable graph remains well below the other pipelines.

A.6 Method-Comparison Summary

Table A1: Population signal and perturbation sources for the compared recovery pipelines under the simplified LOMM block target.

Pipeline	Population signal	Main perturbation	Expected implication
DynEGA	$w^P(\theta_{\text{GLLA}})$: partial readout of GLLA derivatives	Derivative estimation, embedding selection, glasso regularization, finite N, T, community detection	Derivative filter keeps the filtered signal near the shared ceiling
GraphicalVAR	$w^P(\theta_{\text{VAR}})$: partial readout of VAR innovations	VAR estimation, regularization, innovation-projection attenuation, timeouts	Population VAR innovation readout preserves block support under the symmetric block target
GIMME	No closed-form block signal for the implemented readout	Path search, clip-level fitting, extraction, averaging, timeouts	Indirect feasibility-stress comparator

Appendix B: Empirical Realism Numeric Summary

Table A2: Numeric summary for the selected empirical-realism checks. The selected candidate uses a row-wise softmax link with temperature 1.00 and no channel intercepts. Empirical IQR is the scaling denominator for the discrepancy components. Max discrepancy is the larger of the scaled median difference and scaled Wasserstein distance; lower values indicate closer empirical-simulated agreement. Status uses the pass, partial, and fail thresholds defined in the main text.

Readout	Metric	Empirical median	LOMM median	Empirical IQR	Scaled median diff.	Scaled Wasserstein	Max discrepancy	Status
Channels	Channel IQR	0.020	0.064	0.058	0.760	1.058	1.058	Fail
Channels	Derivative IQR	0.014	0.013	0.037	0.007	0.230	0.230	Pass
Channels	Lag-1 ACF	0.521	0.624	0.312	0.331	0.277	0.331	Partial
Channels	Run-length entropy	0.785	0.746	0.132	0.296	0.286	0.296	Pass
Channels	Spectral entropy	0.755	0.685	0.169	0.416	0.500	0.500	Partial
Channels	Effective rank	5.114	5.959	1.347	0.627	0.543	0.627	Partial
Recovered community-score summaries	Channel IQR	0.247	0.832	0.419	1.398	1.481	1.481	Fail
Recovered community-score summaries	Derivative IQR	0.159	0.294	0.258	0.523	0.505	0.523	Partial
Recovered community-score summaries	Lag-1 ACF	0.540	0.745	0.274	0.747	0.564	0.747	Partial
Recovered community-score summaries	Run-length entropy	0.785	0.756	0.132	0.217	0.216	0.217	Pass
Recovered community-score summaries	Spectral entropy	0.751	0.637	0.167	0.686	0.688	0.688	Partial
Recovered community-score summaries	Effective rank	2.371	2.803	0.569	0.758	0.766	0.766	Partial

Appendix C: Conditional Trajectory-Recovery Diagnostics

Table A3: Marginal matched trajectory-recovery diagnostics in the final matched four-indicator grid. Rows use method-readout rows with recovered score summaries and exclude failed or timed-out fits because trajectory summaries exist only after a usable recovered structure is available. Higher trajectory r and lower trajectory NRMSE indicate closer recovery of the generating latent trajectories. Scenario-family rows describe generated activity-and-dynamics regimes, not empirical affect-process categories.

Factor	Level	Usable score-summary rows	Mean trajectory r	Mean trajectory NRMSE
Method	DynEGA	64800	0.725	0.649
Method	GraphicalVAR	62616	0.717	0.659
Method	GIMME	45882	0.604	0.842
Scenario family	A: low-expression single-dominance	32400	0.396	1.167
Scenario family	B: high-expression single-dominance	31791	0.634	0.972
Scenario family	C: high-expression switching	31800	0.581	0.894
Scenario family	D: high-expression rapid switching	30909	0.716	0.686
Scenario family	E: ranked-persistence	23910	0.985	0.118
Scenario family	F: broad switching-geometry	22488	0.999	0.034
Measurement-error SD	0.125	54771	0.814	0.561
Measurement-error SD	0.25	59181	0.704	0.694
Measurement-error SD	0.5	59346	0.562	0.845
Loading strength	0.4	59466	0.631	0.767
Loading strength	0.6	58428	0.703	0.686
Loading strength	0.8	55404	0.740	0.654

Appendix D: Runtime and Feasibility Accounting

Table A4: Runtime and feasibility accounting for the broad DynEGA grid and the matched four-indicator grid. Structure fits count unique fitted structures. Readout rows count the community-detection readouts analyzed in the method-comparison denominator. For GraphicalVAR and GIMME, Louvain, Leiden, and Walktrap are applied to the same fitted structure.

Method	Scope	Structure fits	Readout rows	Valid fits	Timeout fits	Error fits	Valid rows	Timeout rows	Error rows
DynEGA	Matched four-indicator grid	21600	64800	21600	0	0	64800	0	0
DynEGA	Broad DynEGA grid	518400	518400	518400	0	0	518400	0	0
GIMME	Matched four-indicator grid	21600	64800	15294	4114	2192	45882	12342	6576
GraphicalVAR	Matched four-indicator grid	21600	64800	20872	728	0	62616	2184	0

Appendix E: Supplementary Scenario Parameter Values

The final scenario families use the numeric settings below. The reported final runs fix the quality scale at $Q = 0.75$, so the base σ_q vectors in the numeric dynamic-vector table are the final-run innovation vectors, applied as per-step shock standard deviations at the step size used for each series length ($dt = 0.4$ for 100-point series and $dt = 0.25$ for 300-point series). The simulation code also supports configurable reruns with $\sigma_q =$ scenario base $\sigma_q \times (Q/0.75)$.

Table A5: Scenario schedule and feedback-source settings for the final simulation families.

Code	Family	Source	Switching	Gain	Dwell	Feedback source
A	Low-expression single dominance	Generic	No	0.35 / 0.05	1.00 / 1.00	Template
B	High-expression single dominance	Generic	No	1.00 / 0.08	1.00 / 1.00	Template
C	High-expression switching	Generic	Yes	1.00 / 0.08	0.20 / 0.45	Template
D	High-expression rapid switching	Generic	Yes	1.25 / 0.08	0.03 / 0.10	Template
E	Ranked-persistence	Calibration-inspired	No	1.00 / 0.08	1.00 / 1.00	Ranked
F	Broad switching-geometry	Calibration-inspired	Yes	1.10 / 0.10	0.12 / 0.30	Ranked

Table A6: Numeric dynamic vectors for the final simulation families. Base sigma-q values are the final-run values at Q = 0.75.

Code	Dimensions	eta	zeta	Base sigma-q
A	2D	-0.30; -0.55	-1.70; -1.00	0.06; 0.15
A	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.06; 0.15; 0.23
A	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.06; 0.15; 0.23; 0.33
B	2D	-0.30; -0.55	-1.70; -1.00	0.12; 0.30
B	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.12; 0.30; 0.45
B	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.12; 0.30; 0.45; 0.65
C	2D	-0.30; -0.55	-1.70; -1.00	0.12; 0.30
C	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.12; 0.30; 0.45
C	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.12; 0.30; 0.45; 0.65
D	2D	-0.30; -0.55	-1.70; -1.00	0.15; 0.38
D	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.15; 0.38; 0.56
D	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.15; 0.38; 0.56; 0.81
E	2D	-0.05; -0.10	0.08; 0.10	0.25; 0.25
E	3D	-0.05; -0.10; -0.15	0.08; 0.10; 0.12	0.25; 0.25; 0.25
E	4D	-0.05; -0.10; -0.15; -0.20	0.08; 0.10; 0.12; 0.14	0.25; 0.25; 0.25; 0.25
F	2D	-0.05; -0.10	0.08; 0.10	0.35; 0.35
F	3D	-0.05; -0.10; -0.15	0.08; 0.10; 0.12	0.35; 0.35; 0.35

F	4D	-0.05; -0.10; -0.15; -0.20	0.08; 0.10; 0.12; 0.14	0.35; 0.35; 0.35; 0.35